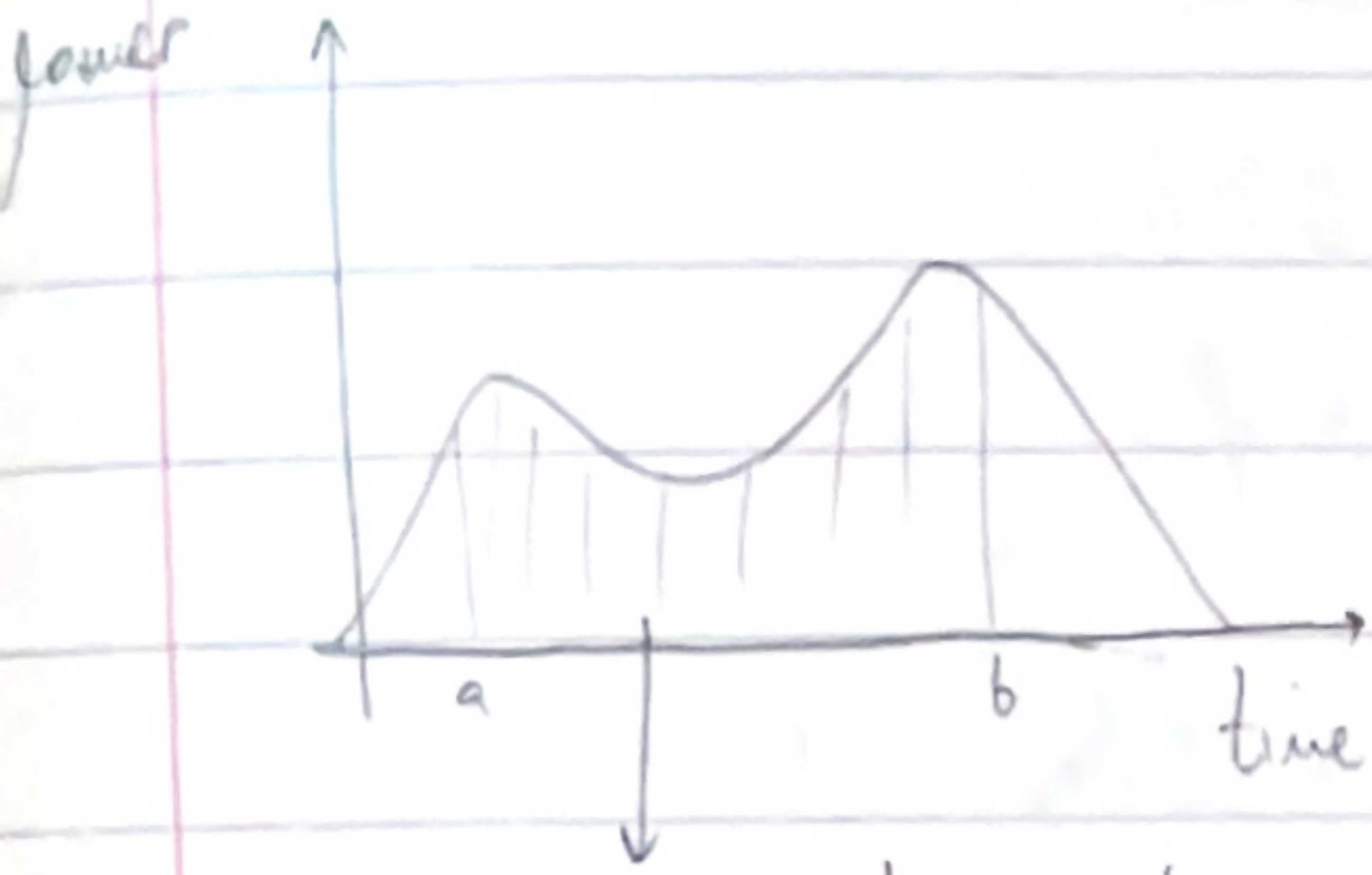
INTEGRATION

Ex. sustainable house



Area debajo de la curva

Integration

- Subintervals of size Δt
- point in each subinterval
- total energy

$$= \underbrace{\sum_{i=1}^{N} P(t_i) \cdot \Delta t}_{\text{Riemann Sum}}$$

- limit $\begin{cases} \Delta t \to 0 \\ N \to \infty \end{cases}$

as $\Delta t \to 0$ → $f(x)$ is called the Integral

$$\int_a^b P(t)\, dt$$

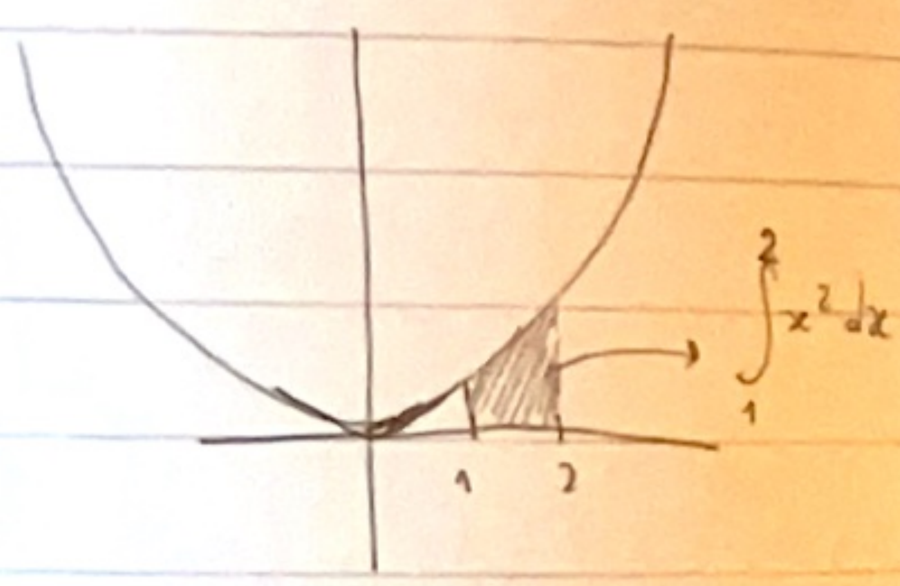
Fundamental Theorem of Calculus

→ f continuous function on $[a, b]$;
→ F function such that $F' = f$;

then $$\int_a^b f(x)\,dx = F(b) - F(a)$$

primitive function or antiderivative of f

Ej.



$f(x) = x^2 \qquad F(x) = \frac{1}{3}x^3$

$F'(x) = \frac{1}{3} \, 3x^2 = x^2$

$$\int_1^2 x^2\,dx = \left[\frac{1}{3}x^3\right]_1^2$$

$$= \frac{1}{3}2^3 - \frac{1}{3}1^3 = \frac{7}{3} \approx 2.33$$

→ Primitive $f(x)$ are never unique (constantes disparaissent dérivées, etc)

Indefinite integral

$$\int f(x)\,d(x) = \text{set of all primitive functions of } f$$

example - $$\int x^2\,dx = \frac{1}{3}x^3 + C$$

Integration : Rules of calculation
Changing the integrand

→ Multiplying by a constant $\int c f(x)\, dx = c \int f(x)\, dx$

→ SUM

Train + man

$$\int f(x) + g(x)\, dx = \int f(x)\, dx + \int g(x)\, dx$$

ex.

$$\int_1^2 4x^3 - 3x^2\, dx = \int_1^2 4x^3\, dx + \int_1^2 -3x^2\, dx$$

$$= 4 \int_1^2 x^3\, dx - 3 \int_1^2 x^2\, dx$$

$$= 4 \left[\frac{1}{4} \cdot x^4\right]_1^2 - 3 \left[\frac{1}{3}\, x^3\right]_1^2$$

$$= 4\left(4 - \frac{1}{4}\right) - 3\left(\frac{8}{3} - \frac{1}{3}\right) = 15 - 7 = 8$$

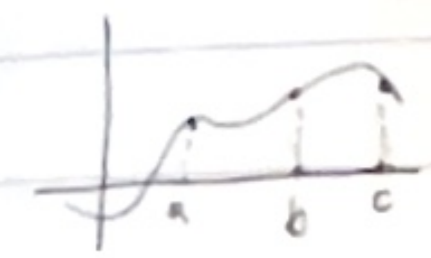
→ Las reglas también funcionan para integrales no definidas

$$\int 4x^3 - 3x^2\,dx = x^4 - x^3 + C$$

→ In integrals = no simple way of calculating $\int f(g(x))\,dx$ or $\int f(x) f(x)$

Splitting an Integral

⇒ Splitting



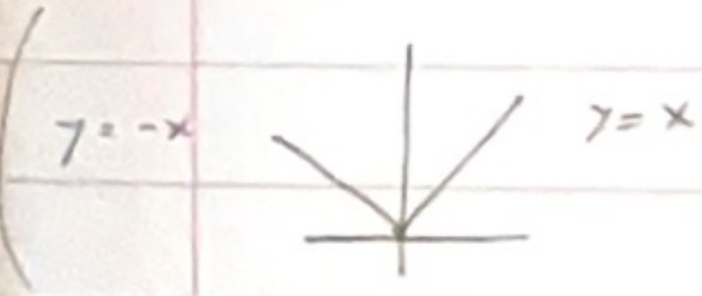
$$\int_a^c f(x)\,dx = \int_a^b f(x)\,dx + \int_b^c f(x)\,dx$$

⇒ Integral with backward interval

$$\int_b^a f(x)\,dx \quad \text{where } b > a$$

$$= -\int_a^b f(x)\,dx$$

$$\int_{-1}^{2} |x|\,dx = \int_{-1}^{0} |x|\,dx + \int_{0}^{2} |x|\,dx = \int_{-1}^{0} -x\,dx + \int_{0}^{2} x\,dx = ⊛$$

(


→ 2 ≠ [illegible] functions)

$$⊛ = \left[\frac{-1}{2}x^2\right]_{-1}^{0} + \left[\frac{1}{2}x^2\right]_{0}^{2} =$$

$$\left(\frac{-1}{2}0^2 - -\frac{1}{2}(-1)^2\right) + \left(\frac{1}{2}2^2 - \frac{1}{2}0^2\right) = \frac{5}{2}$$

The Substitution Method

① ex

$$\int \overbrace{\cos\left(x^3+5x\right)}^{\cos(u)} \overbrace{\left(3x^2+5\right)}^{du} dx \longleftarrow$$

$u = x^3 + 5x$

$\frac{du}{dx} = 3x^2 + 5$

$du = (3x^2+5)\,dx$

$$= \int \cos(u)\,du = \sin(u) + C$$

$$= \sin\left(x^3 + 5x\right) + C$$

$\uparrow$

$u = x^3 + 5x$

ex ②

$$\int \frac{1}{1+x^3}\, x^2 \, dx \longrightarrow$$ Trato de deshacerme de la "inner" function

$u = 1 + x^3$

$\frac{du}{dx} = 3x^2$

$du = 3x^2\,dx$

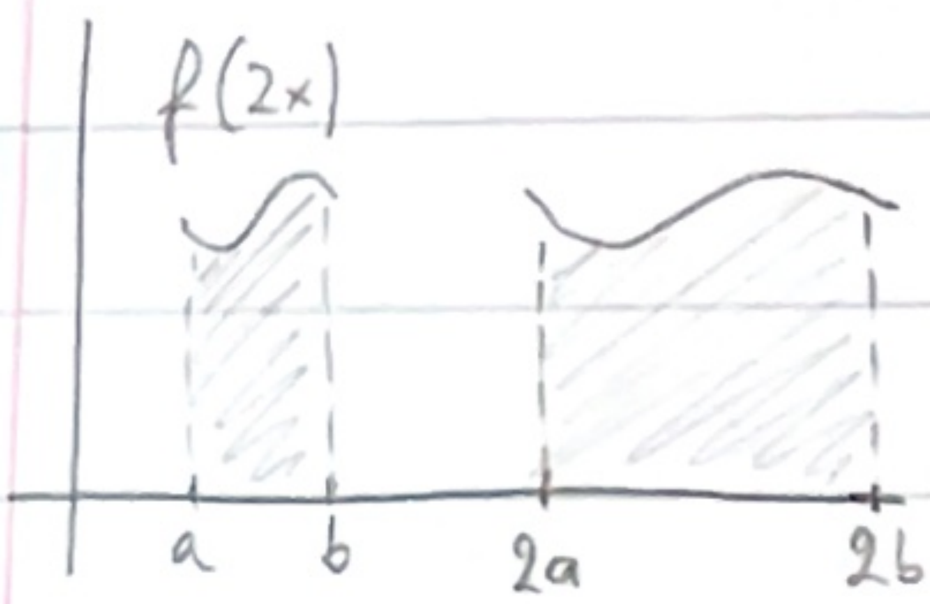
$$= \int \frac{1}{u} \cdot \frac{1}{3}\, du =$$

$$= \frac{1}{3} \ln|u| + C$$

$$= \frac{1}{3} \ln\left|1 + x^3\right| + C$$

Linear Substitutions

linear substitutions → Scaling $x \to Cx$
→ shifting $x \to x + c$



$$\int_a^b f(2x)\,dx = \int_{2a}^{2b} \frac{1}{2} f(x)\,dx$$

↓

$$\int_a^b f(cx)\,dx = \int_{ac}^{bc} \frac{1}{c} f(x)\,dx$$

Tips and tricks

$F' = f \qquad \int f g \, dx = Fg - \int F g' \, dx$

which one to integrate or differentiate?

$$g' \xleftarrow{\text{diff}} g \xrightarrow{\text{int}} \ldots$$

$$1 \longleftarrow x \longrightarrow \frac{1}{2}x^2$$

$$f \qquad\qquad F$$

$$4e^{4x} \longleftarrow e^{4x} \longrightarrow \frac{1}{4}e^{4x}$$

derivative	primitive	
$e^x, \cos(x), -\sin(x)$	$e^x, \sin(x), \cos(x)$	similar
$n x^{n-1}$	x^n	derivar + [illegible]
$\frac{1}{x}$	$\ln(x)$	integrar + [illegible]

Integral of the logarithm

$$\int \ln(x)\,dx = \int 1 \cdot \ln(x)\,dx = x\ln(x) - \int x \cdot \frac{1}{2}\,dx$$

$$= x\ln(x) - x + C$$

"The amazing trick"

int diff

$$\int e^x \cos(x)\,dx = e^x \cos(x) + \int e^x \sin(x)\,dx$$

la hago de nuevo para

$$= e^x \cos(x) + e^x \sin(x) - \int e^x \cos(x)\,dx$$

$$2\int e^x \cos(x)\,dx = e^x \cos(x) + e^x \sin(x)$$

Add

$$\int e^x \cos(x)\,dx = \frac{1}{2}\left(e^x \cos(x) + e^x \sin(x)\right) + C$$